

SECTION C

11 (**short**) Consider the following C++ variables and data structures for representing the positions of pieces on a chessboard at any point.

```
enum PieceType {King, Queen, Bishop, Knight, Rook, Pawn, None};  
enum PieceColour {Black, White};
```

```
class ChessPiece {  
public:  
    PieceType type;  
    PieceColour colour;  
};
```

```
ChessPiece board[8][8]; // 8 x 8 grid of locations
```

Write short segments of C++ code to:

- (a) place a white Rook at a corner of the chessboard (any corner is acceptable); [2]
- (b) print the locations (i.e. row and column) of all the Pawns on the board; [4]
- (c) print the total number of black pieces on the board. [4]